

Hierarchical Flow Diffusion for Efficient Frame Interpolation

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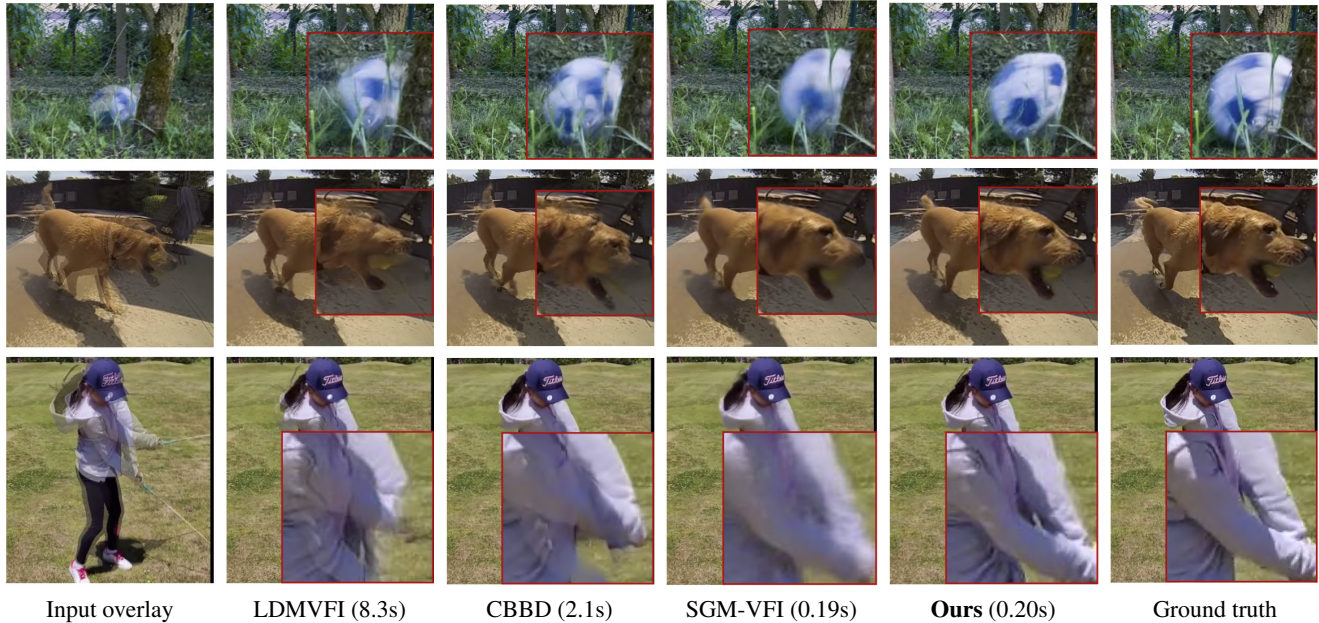


Figure 1. **Different methods for video frame interpolation.** Most diffusion-based [15, 44] interpolation methods (LDMVFI [7], CBBD [32]) still have a large gap from non-diffusion-based methods (SGM-VFI [28]), in both accuracy and efficiency. We propose a diffusion-based model that is 10+ times faster than other diffusion-based methods, and on par with SGM-VFI in efficiency. More importantly, we achieve significantly better accuracy than all baselines. Note how the details and large motions are missed in the baselines, but recovered with our method. We report the inference seconds on the same RTX-4090 GPU with a typical 1024×1024 image pair.

Abstract

Most recent diffusion-based methods still show a large gap compared to non-diffusion methods for video frame interpolation, in both accuracy and efficiency. Most of them formulate the problem as a denoising procedure in latent space directly, which is less effective caused by the large latent space. We propose to model bilateral optical flow explicitly by hierarchical diffusion models, which has much smaller search space in the denoising procedure. Based on the flow diffusion model, we then use a flow-guided images synthesizer to produce the final result. We train the flow diffusion model and the image synthesizer end to end. Our method achieves state of the art in accuracy, and 10+ times faster than other diffusion-based methods. The project page is at: <https://hfd-interpolation.github.io>.

1. Introduction

Video frame interpolation aims to generate intermediate frames given a pair of consecutive frames from video. It is a fundamental video understanding task in computer vision [48], and has many real applications such as slow-motion generation [21], video compression [50], and novel view synthesis [26].

Existing methods have made great progress based on an encoder-decoder paradigm with bilateral flow as an intermediate supervision signal [19, 25, 27, 28, 30, 40, 51, 54]. However, since predicting bilateral flow between two frames is an ill-posed problem with many possible solutions in essence, most of them can only produce an over-smoothed mean solution, as shown in Fig. 1 with SGM-VFI.

Some recent methods try to use diffusion techniques [8, 15, 44] for video frame interpolation [7, 20, 32],

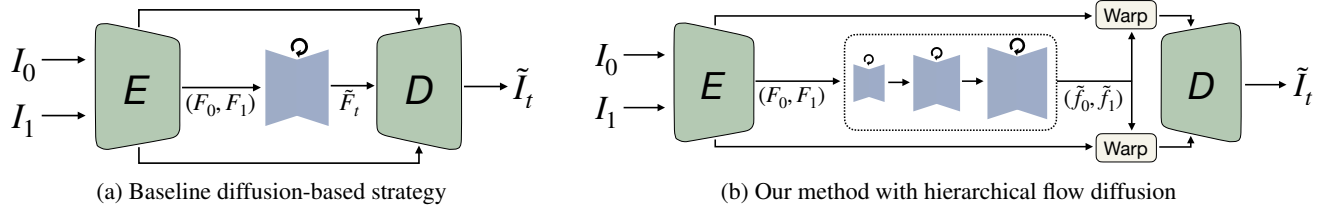


Figure 2. **Different strategies with diffusion models for video frame interpolation.** Given an image pair (I_0, I_1) , our goal is to predict the intermediate frame $\tilde{I}_t$. **(a)** Most diffusion-based methods [7, 20, 32] formulate the problem as a denoising process in the latent space ($\tilde{F}_t$) directly, and train the diffusion network and the encode-decoder (“E” and “D”) network separately. This strategy is less effective caused by the large latent space. On the other hand, this method cannot handle complex motions and large displacement. **(b)** We use a hierarchical strategy with explicit flow modeling. We first train a flow based encoder-decoder for image synthesizer with image pairs and the ground truth optical flow. Then, unlike most diffusion-based methods that denoise the latent space directly, we use a hierarchical diffusion model, conditioned on the encoder feature (F_0, F_1) , to explicitly denoise optical flow from coarse to fine. We use the predicted bilateral flow $(\tilde{f}_0, \tilde{f}_1)$ to warp image features for the synthesizer, and finally fine-tune the synthesizer and the diffusion models jointly.

which formulates the frame interpolation as a denoising process. Although these diffusion-based methods usually generate sharper image results, they suffer from several issues. First, most of them conduct the denoising directly in the latent space, which is less effective because of the large latent space. On the other hand, most of them cannot handle complex motions and large displacement, limited by the representation capability of diffusion networks.

To address these problems, we first train a flow based encoder-decoder for image synthesizer using image pairs and the ground truth optical flow. Then, unlike most diffusion-based methods that model the latent space directly, we use hierarchical diffusion models, conditioned on the encoder feature, to explicitly model optical flow from coarse to fine, which is very efficient and can handle large motions. Finally, we simply upsample the flow as the input of the image synthesizer, and jointly fine-tune the synthesizer and the hierarchical flow diffusion model, as illustrated in Fig. 2.

We evaluate our method on multiple challenging benchmarks, including SNUFILM [6], Xiph [35], DAVIS [39], and Vimeo [52]. Our method achieves state of the art in accuracy, and 10+ times faster than other diffusion-based methods.

2. Related Work

Optical flow estimation is a basic building block for video frame interpolation, and also a fundamental problem in computer vision, which aims to estimate pixel-level matches from the source image to the target image. Traditionally, it is modeled as an energy optimization problem, built upon the assumption of brightness consistency and local smoothness [1, 3, 16, 17]. Recently, learning-based methods have shown great progress, benefiting from large datasets [33, 34, 46] and advanced model architectures [10, 11, 18, 45, 47, 49]. Some recent methods [31, 36, 42] apply diffusion models to optical flow,

and generate promising results. These diffusion models are specifically designed for optical flow estimation, and are trained with ground truth flow. However, in the context of video frame interpolation, we do not have ground truth flow. We propose a hierarchical flow diffusion model supervised by the pseudo flow predicted by a pretrained flow model.

Video frame interpolation has shown significant progress with the development of flow-based methods. Most of them are based on an encoder-decoder paradigm with optical flow as an intermediate supervision signal, in either forward flow supervision [22, 37] or backward flow supervision [19, 25, 27, 40, 54]. The most recent method, SGM-VFI [28], combines the forward-flow and backward-flow techniques in a unified framework, and shows superior performance. However, since the intermediate bilateral flow between two frames has many possible solutions in essence, and we do not have the ground truth bilateral flow for supervision, this type of method tends to produce an over-smoothed mean solution. We propose to only use bilateral flow as intermediate supervisions, and train a flow-guided image synthesizer at the same time, producing an end-to-end trainable system for video frame interpolation.

Diffusion model demonstrates great advantages for many image generation tasks, including text-to-image synthesis [8, 38, 41], image restoration [53, 56], and image editing [13, 23], as well as several high-level tasks, such as monocular depth estimation [24], object detection [4], and image segmentation [5]. Some recent methods [7, 20, 32] apply diffusion models to video frame interpolation and show promising results. However, most of them conduct the denoising procedure in latent space directly, which has a large search space and cannot handle complex motions and large displacement. We propose a hierarchical diffusion model, which formulates the problem as denoising optical flow explicitly in a coarse-to-fine manner, achieving 10+ times faster than other diffusion-based methods.

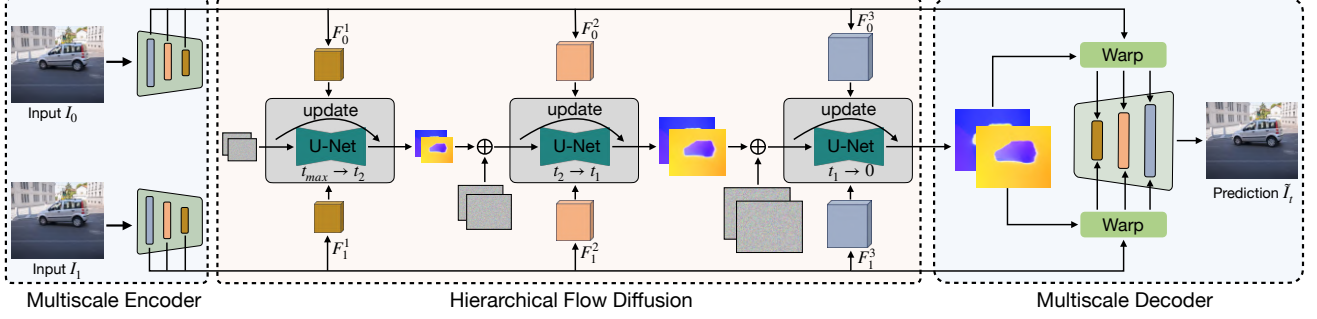


Figure 3. **Overview of our method.** We first construct a flow-guided encoder-decoder with multiscale features as our image synthesizer, and then use diffusion to explicitly denoise optical flow in a coarse-to-fine manner, where the diffusion on each level will be conditioned on encoder features from the corresponding level. With the predicted intermediate optical flow, we use the flow to warp encoder features on each level, and use a multiscale decoder to synthesize the final target image.

3. Approach

Given a pair of consecutive frames, our goal is to generate the intermediate frame. Our method is built on top of the diffusion model [15, 44]. However, unlike common diffusion-based methods, which are usually less efficient and cannot handle complex motions and large displacements, we introduce a hierarchical flow diffusion framework to address those problems. Fig. 3 shows the overview of our framework.

3.1. Flow-Guided Image Synthesis

We train a flow-guided encoder-decoder as our image synthesizer in the first stage. Formally, given a deep regressor g with parameters Φ , let us write

$$\mathbf{I}_t = g(\mathbf{I}_0, \mathbf{I}_1, f_0, f_1; \Phi), \quad (1)$$

where $\mathbf{I}_t$ is the target synthesized image, $\mathbf{I}_0$ and $\mathbf{I}_1$ are the input images, and f_0 and f_1 are the corresponding optical flow from $\mathbf{I}_t$ to $\mathbf{I}_0$ and $\mathbf{I}_1$ respectively. We implement g with a multiscale encoder-decoder architecture [12].

During training, since there is no ground truth bilateral flow (f_0, f_1) in most datasets for frame interpolation. We use a pretrained optical flow network [47] to produce the pseudo bilateral flow $(\tilde{f}_0, \tilde{f}_1)$. We resize the bilateral flow to match the feature resolution on each level, and use the flow to warp encoder features accordingly. After combining the wrapped encoder features and decoder features on each level, the final synthesized result can be inferred from the output of the last decoder layer, as illustrated in Fig. 4. We use 4 channels in the last decoder layer. With one channel for a blending mask M , and three channels for an RGB residual map $\Delta\mathbf{I}$, we have

$$\tilde{\mathbf{I}}_t = M \odot w(\mathbf{I}_0, \tilde{f}_0) + (1 - M) \odot w(\mathbf{I}_1, \tilde{f}_1) + \Delta\mathbf{I}, \quad (2)$$

where w is the warp operation, and $\odot$ is the blending operation. We train the encoder-decoder synthesizer by minimizing the photometric loss [40] between the ground truth target frame $\mathbf{I}_t$ and the prediction $\tilde{\mathbf{I}}_t$. The photometric loss is

a combination of L1-based pixel-wise error $\mathcal{L}_{pixel}$, LPIPS-based perceptual reconstruction error $\mathcal{L}_{lpips}$ [55], and the style loss $\mathcal{L}_{style}$ [9]:

$$\mathcal{L}_{photo} = \mathcal{L}_{pixel} + \lambda_1 \mathcal{L}_{lpips} + \lambda_2 \mathcal{L}_{style}, \quad (3)$$

where λ_1 and λ_2 are balancing parameters.

After finishing the training of the flow-guided encoder-decoder, our hierarchical flow diffusion will be conditioned on features extracted by the pretrained encoder, which will be discussed in the following section.

3.2. Hierarchical Flow Diffusion

We use diffusion to denoise optical flow starting from a Gaussian noise. Unlike most existing diffusion models that are based on only a single fixed resolution, we propose to denoise the optical flow in multiple stages from coarse to fine, as shown in Fig. 5.

With the input image pair $(\mathbf{I}_0, \mathbf{I}_1)$, we exploit the encoder mentioned above to extract the multiscale feature pair

$$\{(\mathbf{F}_0^i, \mathbf{F}_1^i)\}, \quad k_0 \leq i \leq k_1, \quad (4)$$

where the feature level i has a resolution $1/2^i$ of the original image, and k_0 and k_1 denote the finest and coarsest feature level we used in hierarchical diffusion.

At feature level i , our denoising U-Net is conditioned on the feature pair $(\mathbf{F}_0^i, \mathbf{F}_1^i)$, and takes the noisy bilateral flow (f_0^t, f_1^t) as input to predict the target flow $(\tilde{f}_0^i, \tilde{f}_1^i)$, both in the $1/2^i$ of the original resolution. By denoting the denoising U-Net as u with parameter θ , we can write as

$$\tilde{f}_0^i, \tilde{f}_1^i = u(\mathbf{F}_0^i, \mathbf{F}_1^i, f_0^t, f_1^t, t; \theta). \quad (5)$$

Before forwarding the denoising U-Net, we normalize the feature, and use a flow projector and feature projector to process the noisy flow and feature pair, respectively. We share the parameter of the diffusion model across different feature levels, except for the feature and flow projectors.

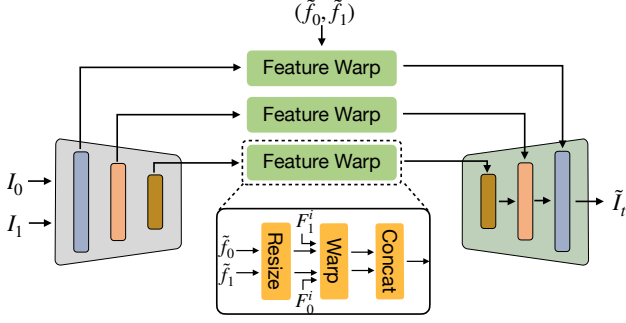


Figure 4. **Illustration of flow-guided image synthesis.** We train a multiscale encoder-decoder as our image synthesizer based on image pairs (I_0, I_1) and bilateral optical flow $(\tilde{f}_0, \tilde{f}_1)$.

In the diffusion procedure, we uniformly divide the entire denoising process $t \sim \mathcal{U}(0, T)$ into multiple stages, while each stage corresponds to a feature level i . We express this as

$$s_t = \{i | t_0^i \leq t < t_1^i\}, \quad t_0^{k_0} = 0, \quad t_1^{k_1} = T, \quad (6)$$

where t_0^i and t_1^i is the starting and ending point, and $(\mathbf{F}_0^i, \mathbf{F}_1^i)$ is used as the condition of the denoising U-Net for interval i .

When time step $t - 1$ and t do not belong to the same denoising stage, i.e., $s_t > s_{t-1}$, we perform $2 \times$ bilinear upsample for the estimated $(\tilde{f}_0^{s_t}, \tilde{f}_1^{s_t})$, and apply the forward function in DDPM [15] to approximate the input for timestep $t - 1$ [56]. We summarize this as

$$f^{t-1} = \sqrt{\alpha_{t-1}} \uparrow \tilde{f}^{s_t} + \sqrt{1 - \alpha_{t-1}} \epsilon, \quad (7)$$

where $\uparrow \tilde{f}^{s_t}$ is $2 \times$ upsampled flow, $\alpha_t \in \{\alpha_1, \dots, \alpha_T\}$ is a predefined noise schedule, and $\epsilon \sim \mathcal{N}(0, 1)$ is the gaussian noise.

When the previous time step $t - 1$ and current time step t falls into the same interval, we update (f_0^{t-1}, f_1^{t-1}) using the reverse function in DDPM [15]

$$f^{t-1} = \sqrt{\alpha_{t-1}} \tilde{f}^{s_t} + \sqrt{1 - \alpha_{t-1} - \sigma_t^2} \tilde{\epsilon}_t + \sigma_t \epsilon, \quad (8)$$

where $\tilde{\epsilon}_t$ is the estimated noise, which is

$$\tilde{\epsilon}_t = (f^t - \sqrt{\alpha_t} \tilde{f}^{s_t}) / \sqrt{1 - \alpha_t}. \quad (9)$$

We apply Eq. 7 and Eq. 8 for $\tilde{f}_0^{s_t}$ and $\tilde{f}_1^{s_t}$ to yield f_0^{t-1} and f_1^{t-1} , respectively.

We train hierarchical diffusion models simultaneously at all feature levels. At each level i , we randomly sample a time step t_i , construct the noisy flow input $(f_0^{t_i}, f_1^{t_i})$ with the resized ground truth flow $(\tilde{f}_0^i, \tilde{f}_1^i)$ matching the resolution of the level i , and apply L1 loss to supervise their prediction $(\tilde{f}_0^i, \tilde{f}_1^i)$

$$\mathcal{L}_{flow} = \sum_{i=k_0}^{k_1} \|\tilde{f}_0^i - f_0^i\|_1 + \|\tilde{f}_1^i - f_1^i\|_1. \quad (10)$$

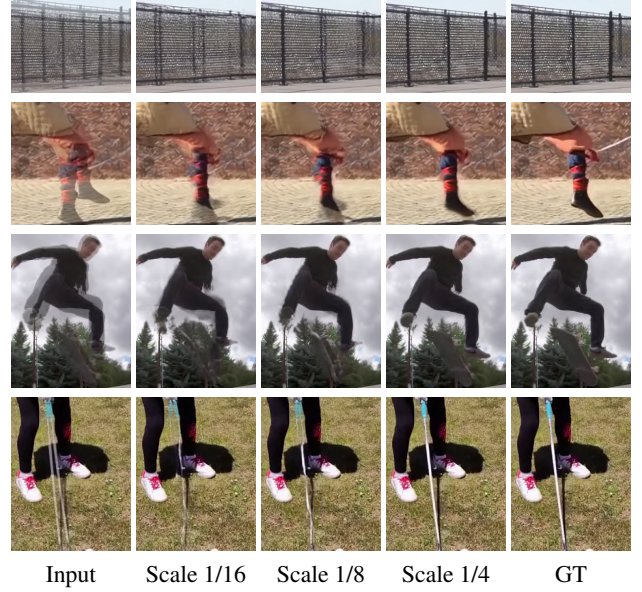


Figure 5. **Results of the hierarchical models on different scales.** We show the coarse-to-fine results from left to right in addition to the input and ground truth. With the proposed hierarchical diffusion models, the result becomes progressively better with finer resolution, making it capable of handling complex motions and large displacements.

3.3. End-to-end Training with Joint Fine-tuning

After finishing the training of the encoder-decoder synthesizer and the hierarchical diffusion models separately, we propose to fine-tune these two components jointly, producing an end-to-end interpolation framework. Given multi-scale bilateral flow $(\tilde{f}_0^i, \tilde{f}_1^i)$ predicted by the hierarchical flow diffusion models, we use them to warp the corresponding features $(\mathbf{F}_0^i, \mathbf{F}_1^i)$ from the encoder to construct the flow-guided features for the decoder. We use the photometric loss in Eq. 3 to supervise the target image generated by the decoder. We will show in experiments that this end-to-end training strategy improves the performance.

4. Experiments

We evaluate our method in this section. We first discuss implementation details of our method and the experimental settings, and then compare it with state-of-the-art methods, followed by systematic ablation studies.

Implementation details. We train our method on Vimeo90k [52], which consists of 50k triplets designed for video frame interpolation. We train the flow diffusion models with the finest resolution of 256×256 . For data augmentation, we randomly crop 256×256 patches and perform random rotation, flipping, and frame order reversing, following previous methods [25, 27, 32]. We use AdamW optimizer [29] and anneal the learning rate from $4e-4$ to $4e-5$ based on One-Cycle strategy [43], for both the training of

Method	easy		medium		hard		extreme	
	LPIPS	FID	LPIPS	FID	LPIPS	FID	LPIPS	FID
AMT-G [27]	0.0325	6.139	0.0447	11.039	0.0680	20.810	0.1128	40.075
SGM-VFI [28]	0.0191	5.854	0.0329	10.945	0.0611	22.004	0.1182	41.078
EMA-VFI [54]	0.0186	5.882	0.0325	11.051	0.0579	20.679	0.1099	39.051
URPNet-LARGE [22]	0.0179	5.669	0.0389	10.983	0.0604	22.127	0.1115	40.098
Per-VFI [51]	0.0166	6.654	0.0263	11.509	0.0480	19.855	0.0901	34.182
LDMVFI [7]	0.0145	5.752	0.0284	12.485	0.0602	26.520	0.1226	47.042
MADIFF [20]	0.0130	5.334	0.0270	11.022	0.0580	22.707	0.1180	44.923
CBBD [32]	<u>0.0112</u>	<u>4.791</u>	<u>0.0274</u>	<u>9.039</u>	<u>0.0467</u>	<u>18.589</u>	<u>0.1040</u>	<u>36.729</u>
Ours	0.0098	4.541	0.0191	8.499	0.0405	15.320	0.0839	27.032

Table 1. **Comparison on SNU-FILM [6] benchmark.** Our method outperforms the current SOTA methods significantly, especially in the hard and extreme subset of SNU-FILM.

Method	2K		4K	
	LPIPS	FID	LPIPS	FID
URPNet-LARGE [22]	0.1010	14.209	0.2150	32.003
EMA-VFI [54]	0.1024	12.332	0.2258	30.675
AMT-G [27]	0.1061	13.089	0.2054	29.512
SGM-VFI [28]	0.1000	12.375	0.2172	27.334
LDMVFI [7]	0.0420	11.385	0.0859	21.272
CBBD [32]	<u>0.0272</u>	<u>10.168</u>	<u>0.0634</u>	<u>24.621</u>
Ours	0.0264	7.940	0.0614	14.132

Table 2. **Comparison on Xiph [35] benchmark.** Our method achieves the best performance across all evaluation settings, especially in the more challenging 4K setting.

Method	DAVIS		Vimeo-90k	
	LPIPS	FID	LPIPS	FID
VFIformer [30]	0.1272	14.407	0.0212	3.341
EMA-VFI [54]	0.1324	15.186	0.0213	3.819
AMT-G [27]	0.1091	13.018	0.0208	3.172
LDMVFI [7]	0.1070	12.554	0.0234	2.744
MADIFF [20]	0.0960	11.089	-	-
Per-VFI [51]	<u>0.0819</u>	8.813	0.0180	2.314
CBBD [32]	0.0919	<u>9.220</u>	<u>0.0123</u>	<u>1.961</u>
Ours	0.0753	7.237	0.0120	1.712

Table 3. **Comparison on DAVIS [39] and Vimeo-90k [52].** Our method outperforms all other competitors, consistently.

the encoder-decoder synthesizer and the diffusion models.

We first train the encoder-decoder synthesizer for 200 epochs with a batch size of 64 and then freeze the synthesizer and train the diffusion model for another 200 epochs with the same batch size. In the final stage, we fine-tune the synthesizer and the diffusion model jointly using the photometric loss discussed in Sec. 3.3 for 100 epochs with a batch size of 32.

We use 3 pyramid levels for hierarchical diffusion models from the coarsest level $k_1 = 4$ to the finest level $k_0 = 2$. The balancing loss weight in photometric loss is set

to $\lambda_1 = 0.1$ and $\lambda_2 = 20$. We use 1000 denoising steps in total for hierarchical diffusion models during training, with the step numbers on each scale roughly the same. To accelerate the denoising process, we follow DDIM [44] to set σ_t in Eq. 8 to 0, formulating it as a deterministic generative process, and perform 6 sampling steps during inference.

During inference, we first resize the input image to a resolution having its shorter dimension equals to 256, and then extract multiscale features, to run the hierarchical diffusion models. After getting the predicted flow, we then resize it to the original image resolution, and feed it into the image synthesizer to get the final interpolated results at the raw resolution.

Evaluation strategy. We train our model only on Vimeo90K, and evaluate it on Vimeo90K and other datasets, including SNUFILM [6] which consists of four subsets (easy, medium, hard, and extreme) with different level of motion magnitude, Xiph [35] which contains 392 triplets in 4K resolution, and DAVIS [39] with 2847 triplets with a fixed resolution of 854×480 . For dataset Xiph, we follow the same preprocessing step as in [37] to generate two datasets, Xiph-4K and Xiph-2K, and evaluate on them, respectively.

We evaluate the interpolated frames in LPIPS [55] and FID [14], which have a better correlation with human perception than PSNR and SSIM [32]. LPIPS calculates the mean squared error (MSE) distance between deep feature embeddings of image pairs, and FID computes the Fréchet distance between the feature distributions of the ground truth and predicted images.

4.1. Comparison to the State of the Art

We compare our method with state-of-the-art methods, including VFIformer [30], AMT [27], SGM-VFI [28], EMA-VFI [54], URPNet [22], and Per-VFI [51]. We also compare our method with the recent diffusion-based methods, including LDMVFI [7], MADiff [20], and CBBD [32]. We report the quantitative results in Table 1, 2, and 3. On

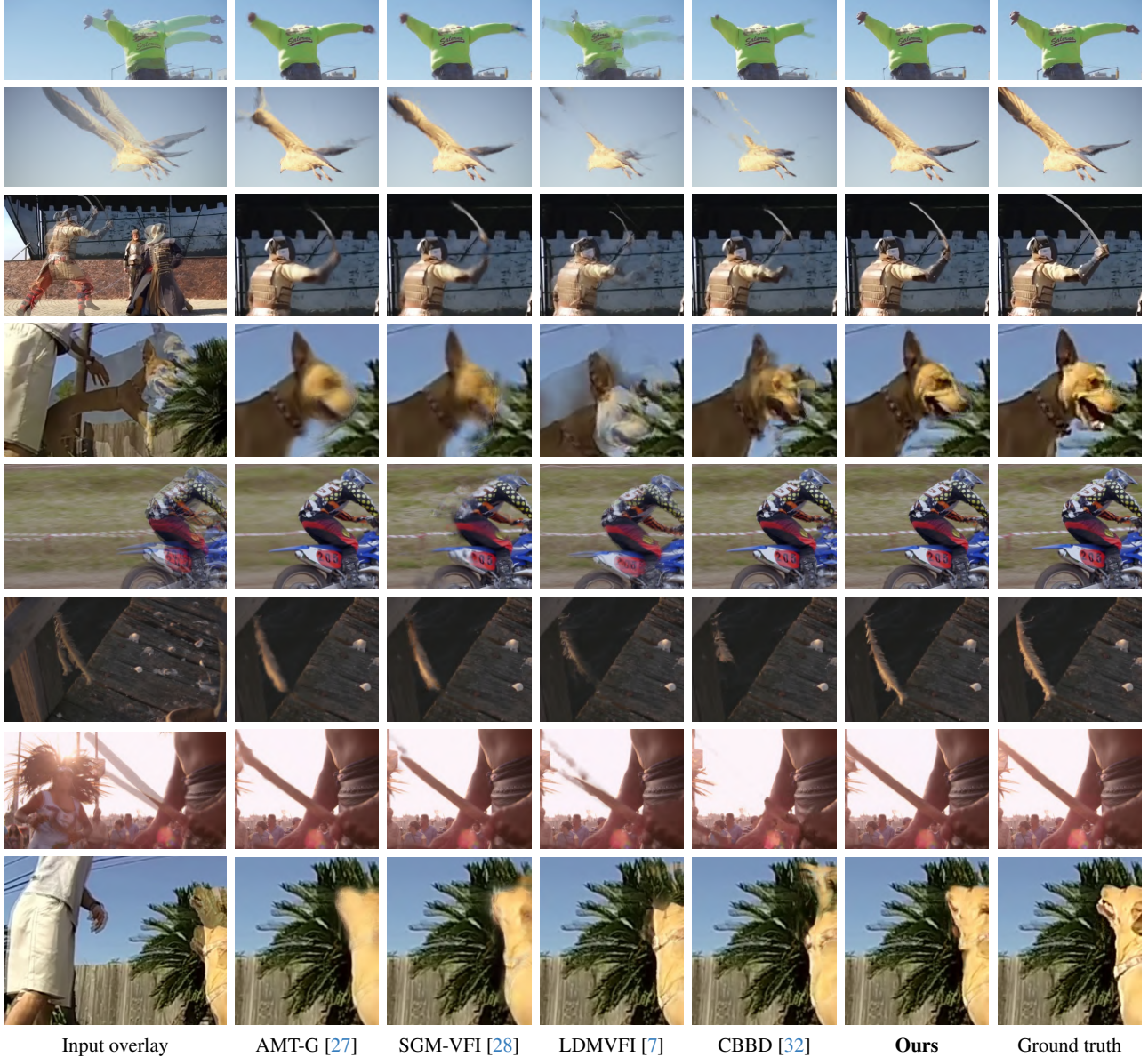


Figure 6. **Qualitative results on SNU-FILM, Xiph and DAVIS.** For complex motions, most non-diffusion-based methods (AMT, SGM-VFI) produce blurry results. However, most diffusion-based methods (LDMVFI, CBBD) struggle in handling large motions. Our method archives the best accuracy and produces high-quality results in most cases, thanks to the proposed hierarchical flow diffusion models.

most datasets, our method outperforms most state-of-the-art methods significantly, especially on more challenging benchmarks such as the extreme subset of SNUFILM and Xiph-4K. Fig. 6 shows qualitative comparison results on SNUFILM, Xiph and DAVIS. Most competitors frequently produce results with blurred results or significant artifacts. By contrast, our method is superior in handling large movements, recovering subtle details, and producing high-quality frame interpolation results.

In addition to the accuracy, we evaluate the efficiency of our method. Since different methods report the running time on different machines in their paper, we run LDMVFI,

CBBD, SGM-VFI, and our method on the same workstation with an RTX-4090 GPU, and set the input image pair at a resolution of 1024×1024 . As discussed in Fig. 1, LDMVFI, CBBD, SGM-VFI, and our method finish the processing in 8.3s, 2.1s, 0.19s, and 0.20s, respectively. Our method is 10+ times faster than the diffusion-based methods LDMVFI and CBBD, and on par with SGM-VFI in efficiency. While note that, our method achieves much higher accuracy than SGM-VFI, as in Table 1 and 2.

Our framework has 0.35M, 0.59M, and 46.96M parameters for the encoder, decoder, and hierarchical diffusion network, respectively, and it only consumes ~ 2.9

Method	SNUFILM-hard		SNUFILM-extreme		Sintel-clean			Sintel-final		
	LPIPS	FID	LPIPS	FID	EPE	LPIPS	FID	EPE	LPIPS	FID
Vanilla	0.0625	22.283	0.1199	46.414	9.19	0.1089	36.079	9.43	0.0939	38.011
Ours	0.0405	15.320	0.0839	27.032	5.70	0.0789	23.476	6.46	0.0730	28.453
Oracle	0.0264	11.050	0.0424	20.710	2.67	0.0427	19.083	3.89	0.0385	21.512
EMA-VFI	0.0579	20.679	0.1099	39.051	6.79	0.1049	28.657	7.28	0.0767	33.948
EMA-VFI + Ours	0.0521	19.144	0.0968	35.346	6.79	0.0863	26.649	7.28	0.0754	29.908

Table 4. **Relation between optical flow and frame interpolation.** “Vanilla” is the setting without our flow diffusion but relying on RAFT to compute the forward and backward flow between the two input frames and multiply by a factor of 0.5 to produce the bilateral flow for the image synthesizer. “Oracle” is the upper bound setting that uses the ground truth interpolation frame to compute the bilateral flow with RAFT. “EMA-VFI + Ours” uses the bilateral flow results of EMA-VFI [54] as the input of our image synthesizer. Our method achieves much better results in both bilateral flow estimation and frame interpolation.

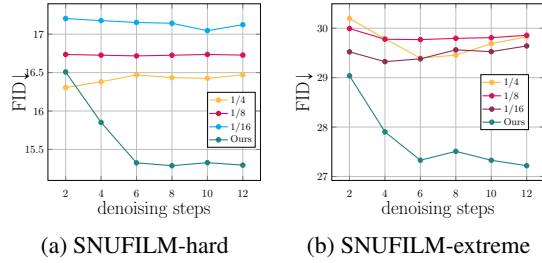


Figure 7. **Analysis of hierarchical diffusion models.** We compare our hierarchical diffusion method with its single-level versions, with which the diffusion only happens at the corresponding level (denoted as “1/16”, “1/8”, and “1/4” respectively). Our hierarchical method improves consistently with more denoising steps.

GB VRAM in float32 mode during inference for a typical 1024x1024 image pair, which is significantly more resource-friendly than other diffusion-based method LD-MVFI (~ 6.9 GB) and CBBB (~ 8.4 GB).

4.2. Ablation Study

Relation between optical flow and frame interpolation.

We first conduct ablation studies on the relation between optical flow and frame interpolation on SNU-FILM. Since there is no ground truth flow in VFI dataset, we also conduct the same ablation on a typical optical flow dataset MPI-Sintel [2] which includes GT flow and two tracks “clean” and “final”. On Sintel, we report the bilateral flow results in end-point error (EPE). As shown in Table 4, “Vanilla” struggles to capture complex motion patterns in estimating bilateral flow and cannot produce reasonable interpolation results. Our method achieves much better results in both bilateral flow estimation and frame interpolation. On the other hand, we compare EMA-VFI and a version with the bilateral flow results of EMA-VFI directly as the input of our image synthesizer (“+ Ours”). With the same bilateral flow, our method is more accurate in frame interpolation, thanks to the proposed image synthesizer.

Result analysis of hierarchical diffusion models. We compare our hierarchical diffusion method with its single-

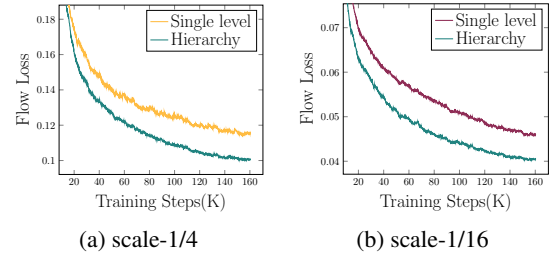


Figure 8. **Training analysis of hierarchical diffusion model.** The proposed hierarchical flow diffusion model has better convergence compared with versions with a fixed resolution on each pyramid level only, which helps to produce better interpolation results with our flow-guided image synthesizer.

level versions, with which the diffusion only happens at the corresponding level. We evaluate their performance with different sampling steps on SUMFILM-hard and SNUFILM-extreme, as illustrated in Fig. 7. Our hierarchical diffusion strategy outperforms all single-level baselines significantly. The single-level baselines struggle to converge, and additional denoising steps bring little improvement. By contrast, our hierarchical diffusion consistently benefits from more denoising steps. This also highlights the robustness of our hierarchical diffusion to different motion types and complex scenes.

Training analysis of hierarchical diffusion models. We report the flow loss value of our hierarchical diffusion model during the training on Vimeo-90k, and compare it with versions with a fixed resolution on a single pyramid level, as shown in Fig. 8. The proposed hierarchical diffusions model converges better than the single-level versions, which helps to produce better interpolation results with our flow-guided image synthesizer.

Effect of flow representation in diffusion. Previous methods [7, 32] use diffusion models to denoise the latent space directly. By contrast, we propose to use diffusion models to denoise an intermediate flow representation. To study the effect of these two parametrization methods within a unified framework, we adapt our encoder-decoder based synthesizer to a latent-based version. Since there is no inter-

Resolution	SNUFILM-hard		SNUFILM-extreme		Runtime					
	LPIPS	FID	LPIPS	FID	Encoder	scale-1/16	scale-1/8	scale-1/4	Decoder	Total
128×128	0.0427	16.737	0.0887	29.774	2.57	17.43	18.48	18.77	7.93	65.31
256×256	0.0405	15.320	0.0842	27.356	2.94	18.23	18.56	18.90	39.31	97.82
512×512	0.0397	15.026	0.0836	26.925	7.45	18.24	18.78	28.64	64.96	138.06

Table 5. **Inference effect of different training resolutions.** We evaluate the inference effect of our model trained with different resolutions. Higher training resolution leads to better performance overall, but resulting in increased inference time. We report the inference time of each stage of our method (in milliseconds), including the different levels in the hierarchical diffusion models (denoted as scale-1/16, scale-1/8, and scale-1/4 respectively). 256×256 gives the best balance between efficiency and accuracy.

Flow	Joint	Share	SNUFILM-hard		SNUFILM-extreme	
			LPIPS	FID	LPIPS	FID
-	✓	✓	0.0442	19.674	0.0956	39.642
✓	-	✓	0.0416	16.158	0.0865	28.767
✓	✓	-	0.0418	16.324	0.0870	29.313
✓	✓	✓	0.0405	15.320	0.0842	27.356

Table 6. **Ablation studies of different design of our method.** “Share” denotes sharing network parameters across hierarchical diffusion models working on different levels, “Joint” denotes jointly fine-tuning the synthesizer and the diffusion model in the last stage, and “Flow” denotes modeling the diffusion process to denoise an optical flow or directly denoise the latent space.

mediate flow, instead of using the warped feature pair, we directly concatenate the encoder features and the decoder feature on corresponding levels. As shown in the 1st and 4th row of Table. 6, without modeling an intermediate flow representation for the encoder-decoder, the performance deteriorates significantly, with the FID error increasing by 28% (from 15.320 to 19.674) on SNUFILM-hard and 45% (from 27.356 to 39.642) on SNUFILM-extreme.

Effect of joint optimization. We jointly optimize the encoder-decoder synthesizer and the diffusion models. We report the result without joint optimization in the 2nd row of Table. 6. The result deteriorates without joint optimization.

Effect of parameter sharing in diffusion models. We share network parameters across the hierarchical diffusion models on different levels. Alternatively, they can use separated network parameters on each level. We evaluate our method in these two settings, and report the result in the last two rows of Table. 6. By sharing network parameters across different levels, we improve the robustness of the diffusion model and decrease the FID error on SNUFILM-hard by 6.6% (from 16.324 to 15.320).

Evaluation of different training resolutions. We train the diffusion model with an input image resolution of 256×256 by default, and for inference, we resize the input image pair to have the shorter size equal to 256 to extract the condition latent. In Table. 5, we evaluate the effect of input image resolution on our method. Increasing the input resolution improves performance. We also list the running time of each component in our framework for running the diffusion



Figure 9. **Limitation discussion.** Our method suffers in scenarios with extreme motion patterns. However, it is still better than the state of the art, and can recover most of the details that are missed with SGM-VFI in these cases.

model on different input image resolutions, and compare it with recent diffusion-based methods in Table. 5. We measure the runtime on a workstation with an NVIDIA RTX-4090 GPU, and as shown in the table, our method takes only 98 ms for 256×256 resolution.

Limitation discussion. Our method produces accurate results in scenarios with complex motion and large displacement, while there are still extreme cases where it produces noticeable artifacts, as shown in Fig. 9. While, note that, our method is still better than the state of the art, and can recover most of the details that are missed with SGM-VFI in those cases. We attribute this to the relatively small-scale training data, and plan to collect datasets in larger scale with diverse motion to further improve the performance.

5. Conclusion

We have introduced a hierarchical flow diffusion model for video frame interpolation. Instead of formulating frame interpolation as a denoising procedure in the latent space, we proposed to model optical flow explicitly from coarse to fine by hierarchical diffusion models, which has much smaller search space in each denoising step, and can handle complex motions and large displacements. In experiments, our approach demonstrates the effectiveness of the hierarchical diffusion models by generating high-quality interpolated frames, which outperforms state-of-the-art methods, and 10+ times faster than other diffusion-based methods.

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